Sprint Review

Surface Crack Detection System

Product Progress & Roadmap

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The Problem & Our Solution

⚠ The Problem

Traditional road inspections are manual, making them:

- Time-consuming and costly
- Prone to human error & difficult to scale

Undetected defects directly lead to:

- Road accidents and safety hazards
- Vehicle damage and high maintenance costs

Our Solution

We are developing an AI-powered Surface Crack Detection System.

The platform enables users to seamlessly upload road surface images and utilizes computer vision to automatically detect, classify, and report the type of surface defect.

Upload → Analyze → Classify → Severity Check

Sprint Goal

This sprint focused on building the foundation of the product by establishing core workflows:

- Developing a secure authentication system
- Building the primary user interface
- Setting up the FastAPI backend services
- Creating an image upload workflow
- Preparing the application structure for seamless AI model integration

SPRINT 1 Progress

Feature / Module	Status
EDA of the dataset	Completed
Authentication Flow (Login, Register, Forgot Password)	Completed
Landing Page & Upload Interface	Completed
Complete UI Design	Completed
FastAPI Backend APIs	Completed
AI Model Training (Initial Version)	Completed
Database Integration & Minor UI Improvements	In Progress

System Workflow

1. Access

User arrives at Landing Page & proceeds to Login or Register.

2. Dashboard

User navigates the Home Page and initiates Image Upload.

3. Processing

System sends image data to the backend AI Model for inference.

4. Insights

Prediction Results are generated, severity levels are checked, and displayed to the user.

Technology Stack

Frontend Interface

Built with **Streamlit** for rapid, data-centric UI development, allowing quick iterations on the dashboard and image upload features.

Backend & Database

Powered by **FastAPI** for high-performance, asynchronous backend routing. Utilizing **Supabase** for robust data storage (in progress).

AI & Deployment

Computer vision models built with **PyTorch + ResNet50**. Version control managed via **Git/GitHub**, targeting **Hugging Face Spaces** for deployment.

Live Demo

This interface is complete and ready for integration with the AI prediction pipeline.

- Landing Page & Registration
- Secure Login Process
- Forgot Password Workflow
- User Dashboard / Home Page
- Upload Image Interface

Product Roadmap

Sprint 2 (Days 5–10)

Goal: Improve AI model and integrate it with the application.

- Enhance dataset using augmentation.
- Compare ResNet50 and EfficientNet-B0.
- Tune model parameters for better accuracy.
- Integrate AI model with the application.
- Complete model evaluation and documentation.

Sprint 3 (Days 11–15)

Goal: Finalize and deploy the product.

- Improve security and UI/UX.
- Perform end-to-end testing.
- Increase model accuracy to ≥90%.
- Deploy on Hugging Face Spaces.
- Complete documentation and project handover.

Current Sprint Progress

The core foundation and UX are fully established. Remaining efforts center on database persistence, end-to-end AI integration, and final testing.

Sprint 1 Review Complete

Thank You!

Open for Questions & Feedback

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